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Studierichting: Informatica
Oefeningenreeks 7D nr. 1.1

$$f(x,y) = x^3 + x^2y^3 - 3y^4$$

$$f(1,-1) = (1)^3 + (1)^2(-1)^3 - 3(-1)^4 = 1 - 1 - 3 = -3$$

$$\begin{cases} \dfrac{\partial f}{\partial x}(x,y) = 3x^2 + 2xy^3 \\ \dfrac{\partial f}{\partial y}(x,y) = 3x^2y^2 - 12y^3 \end{cases} \Rightarrow \begin{cases} \dfrac{\partial f}{\partial x}(1,-1) = 3(1)^2 + 2(1)(-1)^3 = 1 \\ \dfrac{\partial f}{\partial y}(1,-1) = 3(1)^2(-1)^2 - 12(-1)^3 = 15 \end{cases}$$

$$\begin{cases} \dfrac{\partial^2 f}{\partial x^2}(x,y) = 6x + 2y^3 \\ \dfrac{\partial^2 f}{\partial x \partial y}(x,y) = 6xy^2 \\ \dfrac{\partial^2 f}{\partial y^2}(x,y) = 6x^2y - 36y^2 \end{cases} \Rightarrow \begin{cases} \dfrac{\partial^2 f}{\partial x^2}(1,-1) = 6(1) + 2(-1)^3 = 4 \\ \dfrac{\partial^2 f}{\partial x \partial y}(1,-1) = 6(1)(-1)^2 = 6 \\ \dfrac{\partial^2 f}{\partial y^2}(1,-1) = 6(1)^2(-1) - 36(-1)^2 = -42 \end{cases}$$

$$T_2(x,y) = f(a,b) + \frac{\partial f}{\partial x}(a,b)(x-a) + \frac{\partial f}{\partial y}(a,b)(y-b)$$

$$+\frac{1}{2!}\left[\frac{\partial^2 f}{\partial x^2}(a,b)(x-a)^2 + 2\frac{\partial^2 f}{\partial x \partial y}(a,b)(x-a)(y-b) + \frac{\partial^2 f}{\partial y^2}(a,b)(y-b)^2\right]$$

$$\Rightarrow T_2(x,y) = -3+1(x-1)+15(y+1)+\frac{1}{2}\left[4(x-1)^2 + 2(6)(x-1)(y+1) - 42(y+1)^2\right]$$

$$= -3 + (x-1) + 15(y+1) + 2(x-1)^2 + 6(x-1)(y+1) - 21(y+1)^2$$

$$= -14 + 3x - 33y + 2x^2 - 21y^2 + 6xy$$

References